

Practical 9(Trapezoidal Method)

Question 1:

```
In[*]:= a = Input["Enter the left end point:"];
b = Input["Enter the right end point:"];
n = Input["Enter the number of sub-intervals to be formed:"];
h = (b - a) / n;
y = Table[a + i * h, {i, 1, n}];
f[x] := Log[x];
sumodd = 0;
sumeven = 0;
For[i = 1, i < n, i += 2, sumodd += 2 * f[x] /. x -> y[[i]]];
For[i = 2, i < n, i += 2, sumeven += 2 * f[x] /. x -> y[[i]]];
Tn = (h / 2) * ((f[x] /. x -> a) + N[sumodd] + N[sumeven] + (f[x] /. x -> b));
Print["For n=", n, ",Trapezoidal estimate is: ", Tn]
in = Integrate[Log[x], {x, 4, 5.2}];
Print["True value is ", in]
Print["Absolute error is ", Abs[Tn - in]]
```

For n=6,Trapezoidal estimate is: 1.82766

True value is 1.82785

Absolute error is 0.000192256

Question 2:

```

In[*]:= a = Input["Enter the left end point:"];
b = Input["Enter the right end point:"];
n = Input["Enter the number of sub-intervals to be formed:"];
h = (b - a) / n;
y = Table[a + i * h, {i, 1, n}];
f[x] := Sin[x];
sumodd = 0;
sumeven = 0;
For[i = 1, i < n, i += 2, sumodd += 2 * f[x] /. x -> y[[i]]];
For[i = 2, i < n, i += 2, sumeven += 2 * f[x] /. x -> y[[i]]];
Tn = (h / 2) * ((f[x] /. x -> a) + N[sumodd] + N[sumeven] + (f[x] /. x -> b));
Print["For n=", n, ",Trapezoidal estimate is: ", Tn]
in = Integrate[Sin[x], {x, 0, Pi / 2}];
Print["True value is ", in]
Print["Absolute error is ", Abs[Tn - in]]

```

For n=12,Trapezoidal estimate is: 0.998572

True value is 1

Absolute error is 0.0014283

Question 3:

```

In[*]:= a = Input["Enter the left end point:"];
b = Input["Enter the right end point:"];
n = Input["Enter the number of sub-intervals to be formed:"];
h = (b - a) / n;
y = Table[a + i * h, {i, 1, n}];
f[x] := Sin[x] - Log[x] + Exp[x];
sumodd = 0;
sumeven = 0;
For[i = 1, i < n, i += 2, sumodd += 2 * f[x] /. x -> y[[i]]];
For[i = 2, i < n, i += 2, sumeven += 2 * f[x] /. x -> y[[i]]];
Tn = (h / 2) * ((f[x] /. x -> a) + N[sumodd] + N[sumeven] + (f[x] /. x -> b));
Print["For n=", n, ",Trapezoidal estimate is: ", Tn]
in = Integrate[Sin[x] - Log[x] + Exp[x], {x, 0.2, 1.4}];
Print["True value is ", in]
Print["Absolute error is ", Abs[Tn - in]]

```

For n=12,Trapezoidal estimate is: 4.05617

True value is 4.05095

Absolute error is 0.00522485

Question 4:

```

In[* ]:= a = Input["Enter the left end point:"];
b = Input["Enter the right end point:"];
n = Input["Enter the number of sub-intervals to be formed:"];
h = (b - a) / n;
y = Table[a + i * h, {i, 1, n}];
f[x] := 1 / (1 + x^2);
sumodd = 0;
sumeven = 0;
For[i = 1, i < n, i += 2, sumodd += 2 * f[x] /. x -> y[[i]]];
For[i = 2, i < n, i += 2, sumeven += 2 * f[x] /. x -> y[[i]]];
Tn = (h / 2) * ((f[x] /. x -> a) + N[sumodd] + N[sumeven] + (f[x] /. x -> b));
Print["For n=", n, ",Trapezoidal estimate is: ", Tn]
in = Integrate[1 / (1 + x^2), {x, 0, 1}];
Print["True value is ", in]
Print["Absolute error is ", Abs[Tn - in]]

For n=6,Trapezoidal estimate is: 0.784241

True value is  $\frac{\pi}{4}$ 

Absolute error is 0.0011574

```